

ANIL PRADHAN

Full Stack Developer · AI/Agent Systems · Backend Engineering

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[Portfolio](#) · [LeetCode](#) · [GeeksForGeeks](#)

EDUCATION

B.Tech – CSE | CGPA: 9.11/10 · ITER, SOA University, Bhubaneswar 2024 – 2028

Class XII – PCM with IT | 85% · Odisha Board of Secondary Education 2022 – 2024

EXPERIENCE

Campus Ambassador · E-Cell, IIT Kanpur · Remote Dec 2025 – Jan 2026

- Represented IIT Kanpur's E-Summit at ITER, SOA University; boosted student participation via tech talks and networking sessions.

PROJECTS

AI Career Mentor Feb 2026 – Present

Live: [ai-career-mentor-anil.vercel.app](#)

Next.js 14 · FastAPI · LangGraph · ChromaDB · PostgreSQL · Upstash Redis · Groq · NVIDIA NIM · Gemini 2.5 Flash · Docker · GitHub Actions

- Full-Stack AI System Design: Solo-designed and built a production-grade AI career platform orchestrating 6 specialized AI workflows (Resume Parsing, Roadmapping, Market Intelligence, Profile Optimizer, Interview Simulator, and Real-Time Voice Coach) using Next.js 14, FastAPI, Neon PostgreSQL, and Upstash Redis.
- Asynchronous Parallel DAG Orchestration: Architected an async graph execution engine using LangGraph with fan-out/fan-in routing to concurrently parse resumes and scrape market data, reducing cumulative workflow latency by 60%.
- Resilient LLM Routing & Circuit Breakers: Engineered a multi-provider LLM caller layer with custom circuit breakers, exponential backoff retries, and fallback routing chains across Groq, NVIDIA NIM, and Google Gemini for high availability.
- Hybrid RAG & Memory Guards: Implemented a ChromaDB-backed RAG vector retrieval pipeline with in-memory keyword matching fallback, protecting against vector DB connection timeouts and container OOM errors.
- Real-Time WebSockets & Gemini Live Multimodal AI: Built a stateful 7-phase FSM mock interview interface and a full-duplex voice coaching widget (Anya) using Gemini Live Multimodal API for bidirectional 16kHz/24kHz PCM audio streaming.
- Telemetry & Observability: Built an Admin Metrics Console using Redis Sorted Sets (ZSET) for sliding-window session counts, real-time WebSocket tracking, and a background Postgres rollup task (sync_redis_to_postgres) for cost/latency analytics.
- Containerized Deployment & CI/CD: Managed full DevOps lifecycle using Docker multi-stage builds, GitHub Actions (parallel linting, security audits, 104 integration tests), deployed on Render, Vercel, Neon Postgres, and Upstash Redis.

VolunteerIQ Mar 2026 – Apr 2026

Live: [volunteer-iq-clutchcode.vercel.app](#)

Next.js 14 · FastAPI · PostgreSQL · Gemini 2.5 Flash · Groq · Firebase Auth · Mapbox GL · Docker

- Developed backend APIs and AI services for an NGO coordination platform, enabling automated survey analysis, task generation, volunteer matching, and contextual AI-powered assistance.
- Built multi-format document processing pipelines (PDF, CSV, DOCX, TXT) integrated with Gemini AI and Groq fallback mechanisms to ensure reliable humanitarian intelligence workflows.
- Designed secure and scalable backend infrastructure using FastAPI, PostgreSQL, SQLAlchemy, Firebase authentication, Docker, and REST APIs, supporting real-time NGO operations and volunteer management.

TECHNICAL SKILLS

Languages: Java · Python · SQL · JavaScript · TypeScript

Frontend: Next.js 14 · React 18 · Tailwind CSS · shadcn/ui · Lucide React · Recharts · react-dropzone · react-markdown · Axios · react-hot-toast

Backend: FastAPI · Uvicorn · LangGraph · AutoGen/ag2 · LangChain Core · SQLAlchemy 2.0 · Alembic · Pydantic · pdfplumber · edge-tts · SlowAPI · WebSocket · python-jose · bcrypt · sentry-sdk · prometheus-fastapi-instrumentator

AI / LLM & RAG: Groq (Llama 3.3 70B) · NVIDIA NIM · Gemini 2.5 Flash · Gemini Live API · ChromaDB · Serper.dev · Tavily AI · DuckDuckGo

Databases: Neon Postgres · SQLite · Upstash Redis · ChromaDB (Vector Store)

Auth & Security: Google OAuth 2.0 · Firebase Authentication · JWT (python-jose) · bcrypt

DevOps & Infra: Docker · GitHub Actions (CI/CD) · Vercel · Render.com · GHCR · pytest (102 tests)

Tools: VS Code · Git · Swagger UI · Cline · GitHub Copilot · Gemini Code Assist · OpenAI Codex · AntiGravity IDE

ACHIEVEMENTS & CERTIFICATIONS

- Google Cloud Arcade Program (2024–25): Completed with 75 Arcade Points across hands-on GCP labs and challenges.
- GeeksForGeeks 80 Days Coding Challenge: 200+ problems solved across GFG and LeetCode (Arrays, Strings, Recursion, DP, Graphs). Certificate + T-shirt awarded.
- Active in developer communities including GDG and Google tech events, continuously exploring AI systems, cloud infrastructure, backend engineering, and networking with developers through offline meetups and bootcamps.
- Participated in multiple IIT/NIT coding competitions, hackathons, and developer events focused on AI systems, backend engineering, and problem solving.
- Built production-grade AI products across Microsoft AI Dev Days Hackathon (\$80K+), Amazon Nova AI Hackathon (\$95K+), and Google Solution Challenge 2026, transforming ideas into deployable full-stack applications under hackathon timelines.